LINUX PERMISSIONS

Detailed Practical Report

Prepared by	Vivek Satpute
Platform	TryHackMe – TuteDude Cybersecurity Room
Practical objective	Linux navigation, hidden files and file permissions

1. Aim

The aim of this practical was to work with files and directories in Linux, locate normal and hidden files, understand basic permission concepts, and read the contents of **/home/student/practice/restricted_file.txt** on the TryHackMe target machine.

2. Commands used and their explanation

The commands below are explained in the order they appear in the terminal screenshot.

```
pwd
```

Print Working Directory. Displays the full path of the directory in which the terminal is currently working. The output was /home/student.

```
whoami
```

Displays the username of the currently logged-in Linux user. The output was student.

```
ls
```

Lists visible files and folders in the current directory. In the home directory, it showed loot, practice, read_secret.sh, run_secret and secret.txt.

```
cd loot
```

Change Directory. Moves into the folder named loot. The current location becomes /home/student/loot.

```
ls
```

Lists the contents of the current loot folder. It showed readme.txt.

```
cat readme.txt
```

Displays the contents of readme.txt in the terminal. The output was flag{you_know_cd_ls_cat_now}.

```
ls -a
```

Lists all entries, including hidden files. Linux treats filenames beginning with a dot (.) as hidden. This command revealed .hidden_file.txt.

```
cat .hidden_file.txt
```

Displays the contents of the hidden file. The output was flag{you_found_the_hidden_file}.

```
cd ..
```

Moves one level up to the parent directory. From /home/student/loot, it returns to /home/student.

```
ls
```

Lists the visible contents of the home directory again, including the practice folder.

```
cd practice
```

Moves into the practice directory. The current path becomes /home/student/practice.

```
ls
```

Lists the files in the practice directory. It showed restricted_file.txt.

```
cat restricted_file.txt
```

Attempts to display the file contents. In this task, the command successfully displayed flag{you_understood_permissions}.

3. Understanding Linux file permissions

Linux permissions control who can read, modify or execute a file, and who can access a directory. Permissions are assigned to three categories: the owner, the group and other users.

Symbol	Permission	Meaning for a file
r	Read	Allows the user to view the file contents.
w	Write	Allows the user to change the file contents.
x	Execute	Allows the user to run the file as a program or script.
-	No permission	The permission is not granted.

Example: -rw-r----- can be divided into three permission groups after the first character. rw- means the owner can read and write; r-- means the group can read; and --- means other users have no permissions. The first hyphen identifies a regular file.

Useful reference commands: `ls -l` displays detailed file information, including permissions. **`chmod`** changes permissions. For example, **`chmod u+r filename`** adds read permission for the file owner. These are learning examples; the supplied terminal screenshot does not show a chmod command being entered.

4. Results

• File in the loot folder: **readme.txt**
• Contents of readme.txt: **flag{you_know_cd_ls_cat_now}**
• Hidden file: **.hidden_file.txt**
• Contents of hidden file: **flag{you_found_the_hidden_file}**
• Contents of restricted_file.txt: **flag{you_understood_permissions}**

The terminal screenshot records the commands and output. The TryHackMe screenshot shows all four displayed answers marked as correct. The evidence confirms successful reading of the restricted file, but does not show the exact permission-change procedure.

5. Terminal evidence

The screenshot below shows directory navigation, file listing, hidden-file discovery and the contents displayed by the cat commands.

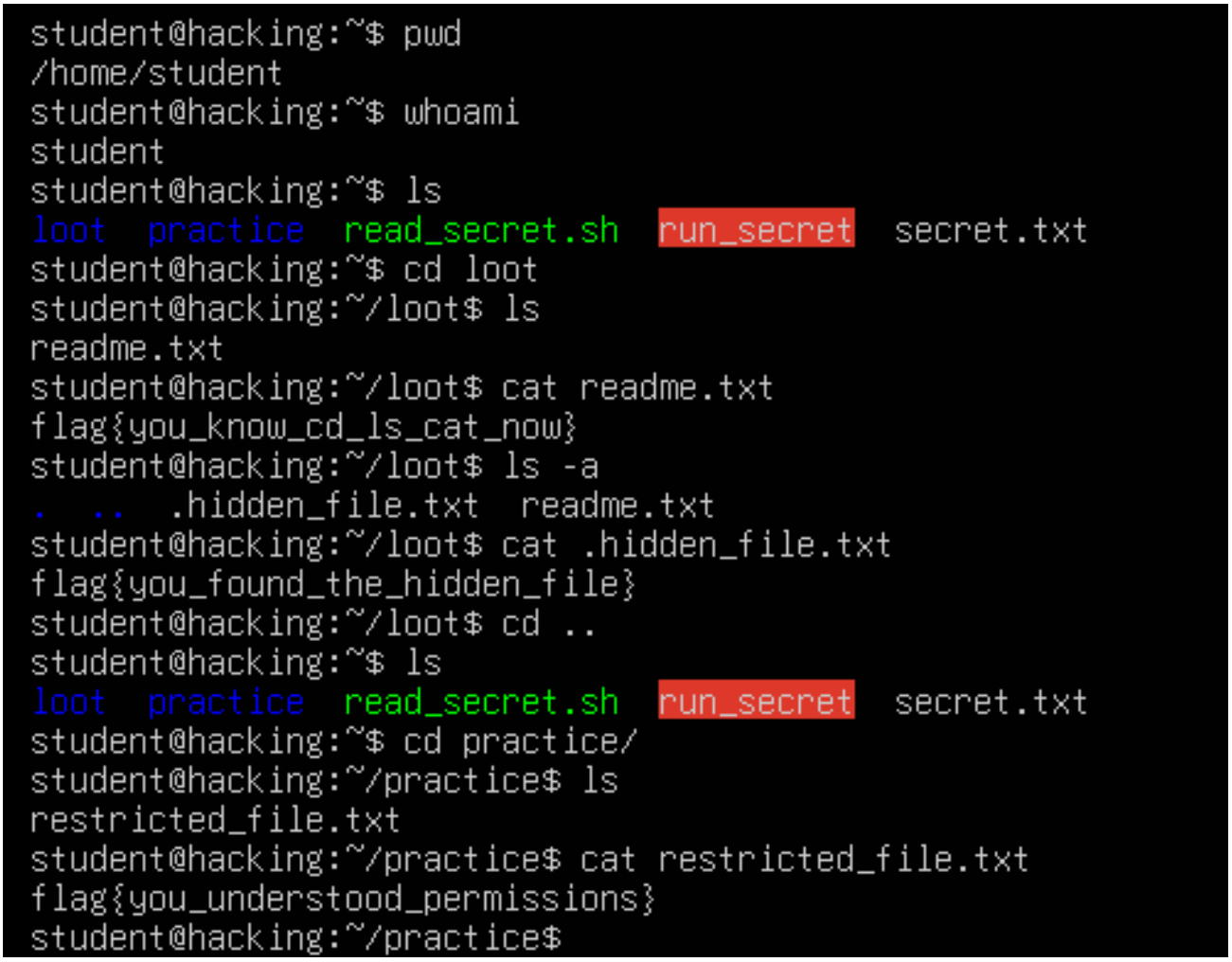


Figure 1. Terminal commands and output from the Linux practice machine.

6. TryHackMe submission evidence

The screenshot below shows the four answers submitted in the TryHackMe room. Each displayed answer is marked “Correct Answer”.

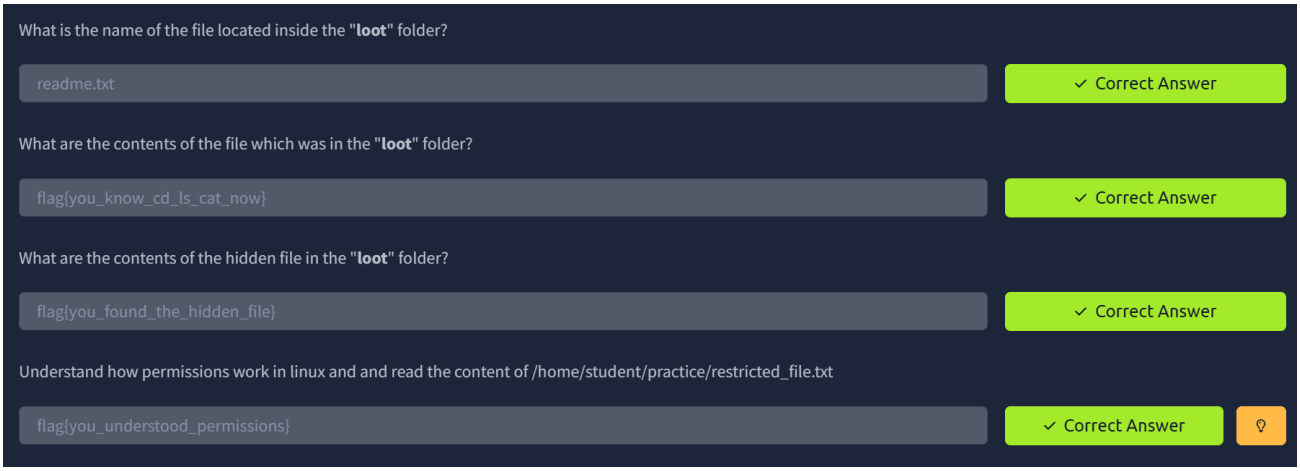


Figure 2. TryHackMe page showing successful answer validation.

7. Conclusion

In this practical, I used Linux commands to check the current directory, identify the logged-in user, list files, move between folders, display file contents and reveal hidden files. I also reviewed the meaning of read, write and execute permissions. The TryHackMe submission page confirms that the answers were accepted.